
Q1.

(a) What is the hexadecimal representation of: `0b1111 1110 1110 1101`?

> **Solution:** Hexadecimal is brilliant, because we are able to directly read each individual nibble as a hexadecimal digit:
>
Binary	Hex	Binary	Hex
0000	0	1000	8
0001	1	1001	9
0010	2	1010	A
0011	3	1011	B
0100	4	1100	C
0101	5	1101	D
0110	6	1110	E
0111	7	1111	F
>
> Thus: 0xFEED

(b) A 32-bit computer has 512 words in its memory. How many individual bytes are there, and what is the required address bus width for this memory?

> **Solution:** Every byte must be addressable. Therefore, each word has four bytes, thus the total number of addressable bytes is $512 \times 4 = 2048$.
>
> The total number of possible addresses for an address bus is known as the address space, and is equal to 2^{width}. Thus, the required width is $log_2(2048) = 11$.
>
> An address bus of width 11 is required, or 11 individual connections.

(c) What is the value of: `(int8_t) ('z' + 42)`?
Hint: This question is broken down into smaller steps in the 'extra' section

> **Solution:** Assuming ASCII, 'z' is encoded in the sequence `0x7A` (not `0x5A` which is 'Z'). The number $42 = (2 \times 16) + (10 \times 1) =$ `0x2A`.
>
> Padding everything to 8 bits:
>
> `0x7A` = 0111 1010
> `0x2A` = 0010 1010
>
> Completing bit-wise addition:
>
	0	1	1	1	1	0	1	0
+	0	0	1	0	1	0	1	0
=	1	0	1	0	0	1	0	0
>
> Now interpreting as an **8-bit signed integer**, we can see that this is negative (as $d_7 = 1$)
>
> We therefore need to $\times -1$ to see the value of the number:

	1	0	1	0	0	1	0	0	↓ invert
	0	1	0	1	1	0	1	1	
+								1	↓ +1
=	0	1	0	1	1	1	0	0	

$= 0 + 64 + 0 + 16 + 8 + 4 + 0 + 0 = 92$

Therefore, the value of `('z' + 42)` interpreted as an 8-bit signed integer is -92.

Q2.

Considering number representation:

(a) What is the 4-bit two's complement representation of -5?

(b) If you treat them as regular 4-bit binary numbers (i.e. ignore the fact that -5 is encoded in two's complement as per above), what is the outcome if you add $7 + (-5)$?

(c) What about $5 + (-5)$?

(d) And $(-5) + 7$?

Solution:

a) Start with $+5 = 4+1 = 0101$, and then multiply by -1. In two's complement, the process to multiply by -1 is to invert all bits, and then add 1:

5_{10}	=	0101
inv:		1010
+		1
	=	1011

b) $7_{10} = 4 + 2 + 1 = 0111$:

		0	1	1	1
+		1	0	1	1
=	(1)	0	0	1	0

c) $5_{10} = 4 + 1 = 0101$:

		0	1	0	1
+		1	0	1	1
=	(1)	0	0	0	0

d) = (1) 0 0 1 0

Q3.

A bar in town is using a new automatic ordering system, where if you display your

drinks order as a *sequence* of lights you get 10% off your bill. You have an array of three LEDs, each of which are able to be set to either (R)ed, (G)reen, or (B)lue.

(a) How many possible patterns are there within this sequence?

(b) If the first LED is used to encode between the (R) Drinks, (G) Meals, and (B) Desserts, how many desserts are able to be encoded?

(c) A new head chef has decided that there needs to be 40 different options for meals. Assuming the type-coding in part b) remains, provide two options for implementing this, one where the number of LEDs stays the same, and one where the number of possible colours stays the same.

(d) Remember that information is encoded within specific What beer is encoded within: (R) (R) (G)?

Code	Meaning
RRR	Weiss
RRG	Stout
RRB	Pilsner
RGR	IPA
RGG	Porter
RGB	Lager
RBR	Munchel
RBG	Session
RBB	APA

Solution:

a): $M^N = 3^3 = 27$ possible sequences.

b): Similar to above, but now we only have 2 LEDs to encode the desserts specifically: $M^N = 3^2 = 9$.

c) $M^{(N-1)} \geq 40$. If $M = 3$ (i.e. the number of colours stays the same), we need to find N such than $3^{(N-1)} \geq 40$, i.e. $N = 5$, or we would need 5 LEDs.

If $N = 3$ (i.e. the number of LEDs stays the same), we need to find M such that $M^{(3-1)} \geq 40$: $M = 7$, or we need 7 different colours.

Q4.

Extra

(a) What is the value of: `(int8_t) ('z' + 42)`?
Let's break it down: The computer will:

- Make a sequence of 1s and 0s to represent `'z'`,
- Make a sequence of 1s and 0s to represent 42,
- Add them together, and
- Interpret this resulting sequence of 1s and 0s (i.e. cast) as a **signed** 8-bit integer.

(a) In either binary or hexadecimal notation, how will the computer represent the character `'z'`?

Solution: Printable chars are stored as ASCII with a look up table. The printable character `'z'` is encoded in the sequence `0x7A` (not `0x5A` which is 'Z').

(b) In binary notation, how will the computer represent the number 42?

Solution: The number $42 = (2 \times 16) + (10 \times 1) =$ `0x2A`.

Step-by-step derivation of the binary representation.
To convert a decimal number to binary by hand, one method is to repeatedly ask: *"What is the largest power of two that still fits inside what remains of our number?"* We place a `1` in that bit position, subtract the power of two from the remainder, and continue with the leftover. If a power of two does *not* fit, we place a `0` in that bit position and move on to the next smaller power.
First, list the powers of two we may need:

bit position n	7	6	5	4	3	2	1	0
2^n	128	64	32	16	8	4	2	1

Now we work from the largest power of two downward, keeping track of the *remainder* at each step.

1. **Start with** 42**.** The largest power of two that fits into 42 is $2^5 = 32$ (since $2^6 = 64 > 42$). So bits 2^7 and 2^6 are `0`, and bit 2^5 is `1`.
$$42 - 32 = 10 \quad \Longrightarrow \quad \texttt{001_____}$$
2. **Remainder** $= 10$**.** Does $2^4 = 16$ fit into 10? No. Bit 2^4 is `0`.
$$10 \quad \Longrightarrow \quad \texttt{0010____}$$
3. **Remainder still** $= 10$**.** Does $2^3 = 8$ fit into 10? Yes. Bit 2^3 is `1`.
$$10 - 8 = 2 \quad \Longrightarrow \quad \texttt{00101___}$$
4. **Remainder** $= 2$**.** Does $2^2 = 4$ fit into 2? No. Bit 2^2 is `0`.
$$2 \quad \Longrightarrow \quad \texttt{001010__}$$
5. **Remainder still** $= 2$**.** Does $2^1 = 2$ fit into 2? Yes. Bit 2^1 is `1`.
$$2 - 2 = 0 \quad \Longrightarrow \quad \texttt{0010101_}$$
6. **Remainder** $= 0$**.** Nothing left. Bit 2^0 is `0`.
$$0 \quad \Longrightarrow \quad \texttt{00101010}$$

Collecting the bits from most-significant to least-significant:

$$42_{10} \;=\; \texttt{0010 1010}_2 \;=\; \texttt{0x2A}.$$

Sanity check by expanding the binary back out:

$$\underbrace{0 \cdot 128}_{0} + \underbrace{0 \cdot 64}_{0} + \underbrace{1 \cdot 32}_{32} + \underbrace{0 \cdot 16}_{0} + \underbrace{1 \cdot 8}_{8} + \underbrace{0 \cdot 4}_{0} + \underbrace{1 \cdot 2}_{2} + \underbrace{0 \cdot 1}_{0} = 42. \;\checkmark$$

Grouping the eight bits into two nibbles of four gives `0010` = 2 and `1010` = 10 = `A`, which reproduces the hexadecimal form `0x2A` stated above.

AI Use Disclaimer. The step-by-step derivation of the binary representation of 42 shown above was generated with the assistance of a large language model (Anthropic *Claude Opus 4.7*, accessed 17 July 2026). The model was prompted to produce a "beautifully slow, step-by-step instructional answer" that determines the binary representation of 42 by repeatedly identifying the largest power of two that fits into the current remainder, subtracting it, recording the corresponding bit, and iterating until the remainder reaches zero, with the output formatted in LaTeX to sit beneath the existing solution. The generated content was reviewed and verified by the author prior to inclusion.

(c) What is the result of adding (a) and (b) together?
Hint: This is most easily done in binary column-by-column addition.

Solution: Padding everything to 8 bits:
`0x7A` = 0111 1010
`0x2A` = 0010 1010
Completing bit-wise addition (note the top row is the carry from each column's respective right-hand neighbouring column):

	0	1	1	1	1	0	1	0
+	0	0	1	0	1	0	1	0
=	1	0	1	0	0	1	0	0

(d) How are all signed (i.e. positive or negative) integers stored in computers?

Solution:
Two's complement.

(e) What is the value therefore stored in the sequence of 1s and 0s from (c), interpreted as an 8-bit signed integer?

Solution: Now interpreting as an **8-bit signed integer**, we can see that this is negative (as $d_7 = 1$)

We therefore need to $\times -1$ to see the value of the number:

	1	0	1	0	0	1	0	0	↓ invert
	0	1	0	1	1	0	1	1	
+								1	↓ +1
=	0	1	0	1	1	1	0	0	

$= 0 + 64 + 0 + 16 + 8 + 4 + 0 + 0 = 92$

Therefore, the value of `('z' + 42)` interpreted as an 8-bit signed integer is -92.

(f) Extra for experts: What is the 32-bit floating point representation of -23.25?

Solution: -23.25 in binary is $(16) + (4) + (2) + (1) + (\frac{1}{4})$

Or:

$$\begin{aligned} -23.25_{10} &= -10111.01_2 \\ &= -1.011101 \times 10^{100} && \text{(With everything in base 2)} \\ &= -1.011101 \times 10^{(10000011-01111111)} \end{aligned}$$

Thus:

S $= 1$

E $= 1000\ 0011$ (Exponent is $E - 127$)

F $= 01110100..00$ (23 bits in total)

The total sequence is:

31 0

1	1	0	0	0	0	0	1	1	0	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

END OF QUESTIONS